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ROTATIONALLY SYMMETRIC FOLDED FLANGE PAPER FOOD CONTAINERS

Abstract

Disclosed is a container that has a base, a body, and a crown. The base includes a base fold. The body is circumferentially connected to the base and includes at least one face section and at least one fold section. The crown is circumferentially connected to the base and includes at least one face section and at least one fold section.

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Background/Summary

[0001] This application claims the priority and benefit of U.S. Provisional Patent Application No. 63/559,426 filed on Feb. 29, 2024, which is incorporated by reference in its entirety.

BACKGROUND

[0002] Paper containers have a long history dating back to the 19th century, primarily as solutions for carrying and storing dry goods. Early innovations such as the square-bottom paper bag provided a lightweight, cost-effective alternative to heavy glass or metal containers, making it easier to package and transport various goods. These containers offered advantages such as biodegradability, affordability, and easy manufacturing. However, as demand for packaged food and beverages grew in the 20th century, new challenges emerged, particularly around durability, moisture resistance, and ease of storage.

[0003] The evolution of paper containers led to significant innovations in multi-layered constructions and coated linings, improving their strength and resistance to moisture and grease. Examples include wax-coated paper cups for beverages and the gable-top carton for milk. While these advancements expanded the applications for paper containers, many traditional designs still relied on rigid shapes, which can limit their practicality and storage efficiency, especially in the retail and food service industries.

[0004] Foldable paper containers emerged as an effective solution to these limitations. By allowing the container to be easily collapsed and reassembled, foldable designs enhance space efficiency in storage, shipping, and disposal. Furthermore, foldable paper containers often require less material to manufacture than rigid alternatives, making them more environmentally sustainable and cost-effective. The ability to compactly fold these containers also reduces transportation costs and carbon emissions, aligning with the increasing demand for eco-friendly packaging solutions.

[0005] Despite these advantages, challenges remain in optimizing foldable paper containers to balance durability, leak-proofing, and ease of use. Innovations in foldable container designs and materials that enhance their strength and barrier properties, while maintaining their collapsible nature, represent a valuable advancement in the field of sustainable packaging. This invention addresses these needs, providing a foldable paper container with improved structural integrity, and ease of use.

SUMMARY OF THE DISCLOSURE

[0006] Disclosed herein is a rotational symmetric folded flanged container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Non-limiting and non-exhaustive implementations of the disclosure are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified. The advantages of the disclosure will become better understood with regard to the following description and accompanying drawings where:

[0008] FIG. **1** illustrates a top view of a rotational symmetric unfolded flanged container.

[0009] FIG. **2** illustrates a perspective view of a rotational symmetric folding flanged container.

[0010] FIG. **3** illustrates a perspective view of a rotational symmetric folding flanged container.

[0011] FIG. **4** illustrates a perspective view of a rotational symmetric folding flanged container.

[0012] FIG. **5** illustrates a perspective view of a rotational symmetric folded flanged container.

DETAILED DESCRIPTION

[0013] In the following description of the disclosure, reference is made to the accompanying drawings, which form a part hereof, and which are shown by way of illustration-specific implementations in which the disclosure may be practiced. It is understood that other implementations may be utilized, and structural changes may be made without departing from the scope of the disclosure.

[0014] In the following description, for purposes of explanation and not limitation, specific techniques and embodiments are set forth, such as particular techniques and configurations, in order to provide a thorough understanding of the device disclosed herein. While the techniques and embodiments will primarily be described in context with the accompanying drawings, those skilled in the art will further appreciate that the techniques and embodiments may also be practiced in other similar devices.

[0015] Reference will now be made in detail to the exemplary embodiments, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used throughout the drawings to refer to the same or like parts. It is further noted that elements disclosed with respect to particular embodiments are not restricted to only those embodiments in which they are described. For example, an element described in reference to one embodiment or figure may be alternatively included in another embodiment or figure regardless of whether or not those elements are shown or described in another embodiment or figure. In other words, elements in the figures may be interchangeable between various embodiments disclosed herein, whether shown or not.

[0016] FIG. **1** illustrates a top view of rotational symmetric unfolded flanged container **100**. Container **100** may include a base **105**, body **110**, and crown **115**. Base **105** may be circumferentially connected to body **110** by base fold **120**. The shape of base fold **120** forms a circle. Alternatively, base fold **120** may be a pentadecagon or other polygonal shape depending on the number of sections **140**A-N and **145**A-N. Body **110** may be connected to crown **115** by body fold **125**. Body fold **125** forms a pentadecagon or other polygonal shape depending on the number of sections **140**A-N and **145**A-N. Body **110** may include multiple face sections **140**(A-N) and multiple fold sections **145**A-N. For example, body **110** may include fold section **145**B positioned between face section **140**B and fold section **140**C. In this manner, body **110** may alternate between face section **140**A, fold section **145**A, to face section **140**B, and fold section **145**B, to face section **140**C, and fold section **145**C, to face section **140**N, and fold section **145**N and so on. This process may extend circumferentially alternating face sections **140**N and fold sections **145**N on body **110**. Face sections **140**A-N and fold sections **145**A-N may touch base **105** at a proximal end and may touch crown **115** at a distal end. Proximal in this disclosure means closer to the center and distal in this disclosure means closer to an outside edge.

[0017] Face sections **140**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Similarly, fold sections **145**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Face sections **140**A-N may consist of a trapezoid-like shape with a curved bottom towards the proximal end where it meets base fold **120**. Alternatively, face section **140**A-N may be a trapezoid. Fold sections **145**A-N are smaller than half the size of face sections **140**A-N but also resemble a trapezoid-like shape with a curved side. The curved side is located on the proximal end of fold sections **145**A-N where it meets base fold **120**. Alternatively, fold section **145**A-N may be a trapezoid. The size and shape of face sections **140**A-N and **150**A-N and fold sections **145**A-N and **155**A-N may vary based on the desired shape of container **100**.

[0018] Folds **120**, **125**, **130**A-N, and **135**A-N may indicate where a bend in the material may occur or near where a bend is intended to be. Further, folds **120**, **125**, **130**A-N, and **135**A-N may represent a score in the material to facilitate bending in or near that location. Some bends may occur right along folds **120**, **125**, **130**A-N, and **135**A-N. In another example, a bend may occur near or around an area. For example, base fold **120** may be circularly shaped or circularly scored prior to the folding process to facilitate the fold. However, when the process of folding takes place the shape of the base fold **120** may resemble more polygonal than circular. Container **100** may be comprised of one or more of wood, paper, cardboard, metal, (aluminum, copper, brass), plastic, foam, rubber, or other foldable material known in the art.

[0019] Once folded, container **100** may include multiple resting positions where material forces facilitated container **100** to remain in that position. For example, one position would be the flat position prior to fold initiation. A second example may be when container **100** is folded together such that a small polygonal-shaped opening is displayed at the top portion of the container and where the polygonal-shaped opening is positioned above body fold **125**. The third example may be when container **100** is folded together such that the polygonal-shaped opening is positioned below body fold **125**.

[0020] As container **100** is folded face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a counterclockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a clockwise direction. Facilitating the production of multiple resting positions. In an alternative embodiment, face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a clockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a counterclockwise direction.

[0021] Container **100** may be folded along base fold **120** towards base **105** such that base **105** may act as the base of container **100** and body **110** may act as the walls of container **100**. The angle of body **110** with respect to base **105** may vary. A greater angle between base **105** and body **110** creates a more gradual slope in the wall of container **100** wherein the distal end of body **110** extends away from a center point of container **100**. As the angle between body **110** and base **105** decreases the distal ends of body **110** come closer to a center point of container **100**. Accordingly, body **110**, acting like the walls of container **100**, becomes steeper. As a result, body **110** may be folded such that it may be substantially perpendicular to the plane of base **105**. Substantially perpendicular in this circumstance means plus or minus 10 degrees.

[0022] Crown **115** may be circumferentially attached to body **110** and may fold inward at base fold **120** towards the center point of container **100**. Crown **115** may be folded at an angle such that the distal ends of crown **115** come together and extend over the top of base **105** to form a small polygonal-shaped opening. This opening, as seen in FIG. **5**, may be positioned at the top portion of the container such that the opening is above (or higher than) a plane defined by body fold **125** creating a resting position for container **100**. Alternatively, the opening, positioned over base **105**, may be even with or below the plane defined by body fold **125** to create another resting position for container **100**.

[0023] To further facilitate the shape and resting positions of container **100**, container **100** may be folded along the one or more spoke folds **130**A-N and angled spoke folds **135**A-N. One or more spoke folds **130**A-N may fold towards the inside portion of container **100** on body **110**. At the same time, the same one or more spoke folds **130**A-N may be folded towards the outside portion of container **100** on the crown **115**. At the same time, one or more angled spoke folds **135**A-N may fold towards the outside portion of container **100** on body **110**. At the same time, the same one or more angled spoke folds **135**A-N may be folded towards the inside portion of container **100** on the crown **115**. In short, on body **110**, spoke folds **130**A-N may be folded in opposite directions as compared to angled spoke folds **135**A-N. On crown **115**, spoke folds **130**A-N may be folded in the opposite direction compared to spoke folds **130**A-N on body **110**. Similarly, on crown **115** angled spoke folds **135**A-N may be folded in opposite directions compared to angled spoke folds **135**A-N on body. As a result, following this pattern, regardless of the direction of the folds outward or inward, may create multiple resting positions for container **100**. The elasticity of the material of container **100** and the friction created by folds **120**, **125**, **130**A-N, and **135**A-N may facilitate the resting positions of container **100**.

[0024] FIG. **2** illustrates a perspective view of a rotational symmetric folding flanged container. Container **100** may include a base **105**, body **110**, and crown **115**. Base **105** may be circumferentially connected to body **110** by base fold **120**. The shape of base fold **120** forms a circle. Alternatively, base fold **120** may be a pentadecagon or other polygonal shape depending on the number of sections **140**A-N (**140**A is not seen due to perspective) and **145**A-N (sections **145**A is not seen due to perspective). Body **110** may be connected to crown **115** by body fold **125**. Body fold **125** forms a pentadecagon or other polygonal shape depending on the number of sections **140**A-N and **145**A-N. Body **110** may include multiple face sections **140**(A-N) and multiple fold sections **145**A-N. For example, body **110** may include fold section **145**B positioned between face section **140**B and fold section **140**C. In this manner, body **110** may alternate between face section **140**A, fold section **145**A, to face section **140**B, and fold section **145**B, to face section **140**C, and fold section **145**C, to face section **140**N, and fold section **145**N and so on. This process may extend circumferentially to where face sections **140**A-N and fold sections **145**A-N touch base **105** at a proximal end and touch crown **115** at a distal end. Proximal in this disclosure means closer to the center and distal in this disclosure means closer to an outside edge.

[0025] Face sections **140**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Similarly, fold sections **145**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Face sections **140**A-N may consist of a trapezoid-like shape with a curved bottom towards the proximal end where it meets base fold **120**. Alternatively, face section **140**A-N may be a trapezoid. Fold sections **145**A-N are smaller than half the size of face sections **140**A-N but also resemble a trapezoid-like shape with a curved side. The curved side is located on the proximal end of fold sections **145**A-N where it meets base fold **120**. Alternatively, fold section **145**A-N may be a trapezoid. The size and shape of face sections **140**A-N and **150**A-N and fold sections **145**A-N and **155**A-N may vary based on the desired shape of container **100**.

[0026] Folds **120**, **125**, **130**A-N, and **135**A-N may indicate where a bend in the material may occur or near where a bend is intended to be. Further, folds **120**, **125**, **130**A-N, and **135**A-N may represent a score in the material to facilitate bending in or near that location. Some bends may occur right along folds **120**, **125**, **130**A-N, and **135**A-N. In another example, a bend may occur near or around an area. For example, base fold **120** may be circularly shaped or circularly scored prior to the folding process to facilitate the fold. However, when the process of folding takes place the shape of the base fold **120** may resemble more polygonal than circular. Container **100** may be comprised of one or more of wood, paper, cardboard, metal, (aluminum, copper, brass), plastic, foam, rubber, or other foldable material known in the art.

[0027] Once folded container **100** may include multiple resting positions where material forces facilitated container **100** to remain in that position. For example, one position would be the flat position prior to fold initiation. A second example may be when container **100** is folded together such that a small polygonal-shaped opening is displayed at the top portion of the container and where the polygonal-shaped opening is positioned above body fold **125**. The third example may be when container **100** is folded together such that the polygonal-shaped opening is positioned below body fold **125**.

[0028] As container **100** is folded face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a counterclockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a clockwise direction. Facilitating the production of multiple resting positions. In an alternative embodiment, face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a clockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a counterclockwise direction.

[0029] Container **100** may be folded along base fold **120** towards base **105** such that base **105** may act as the base of container **100** and body **110** may act as the walls of container **100**. The angle of body **110** with respect to base **105** may vary. A greater angle between base **105** and body **110** creates a more gradual slope in the wall of container **100** wherein the distal end of body **110** extends away from a center point of container **100**. As the angle between body **110** and base **105** decreases the distal ends of body **110** come closer to a center point of container **100**. Accordingly, body **110** acts like the walls of container **100** become steeper. As a result, body **110** may be folded such that it may be substantially perpendicularly to the plane of base **105**. Substantially perpendicular in this circumstance means plus or minus 10 degrees.

[0030] Crown **115** may be circumferentially attached to body **110** and may fold inward at base fold **120** towards the center point of container **100**. Crown **115** may be folded at an angle such that the distal ends of crown **115** come together and extend over the top of base **105** to form a small polygonal-shaped opening. This opening, as seen in FIG. **5**, is positioned at the top portion of the container and where the opening is above (or higher than) a plane defined by body fold **125** creating a resting position for container **100**. Alternatively, the opening, positioned over base **105**, may be even with or below the plane defined by body fold **125** to create another resting position for container **100**.

[0031] To further facilitate the shape and resting positions of container **100**, container **100** may be folded along the one or more spoke folds **130**A-N and angled spoke folds **135**A-N. One or more spoke folds **130**A-N may fold towards the inside portion of container **100** on body **110**. At the same time, the same one or more spoke folds **130**A-N may be folded towards the outside portion of container **100** on the crown **115**. At the same time, one or more angled spoke folds **135**A-N may fold towards the outside portion of container **100** on body **110**. At the same time, the same one or more angled spoke folds **135**A-N may be folded towards the inside portion of container **100** on the crown **115**. In short, on body **110**, spoke folds **130**A-N may be folded in opposite directions as compared to angled spoke folds **135**A-N. On crown **115**, spoke folds **130**A-N may be folded in the opposite direction compared to spoke folds **130**A-N on body **110**. Similarly, on crown **115** angled spoke folds **135**A-N may be folded in opposite directions compared to angled spoke folds **135**A-N on body. As a result, following this pattern, regardless of the direction of the folds outward or inward, may create multiple resting positions for container **100**. The elasticity of the material of container **100** and the friction created by folds **120**, **125**, **130**A-N, and **135**A-N may facilitate the resting positions of container **100**.

[0032] FIG. **3** illustrates a perspective view of a rotational symmetric folding flanged container. Container **100** may include a base **105**, body **110**, and crown **115**. Base **105** may be circumferentially connected to body **110** by base fold **120**. The shape of base fold **120** forms a circle. Alternatively, base fold **120** may be a pentadecagon or other polygonal shape depending on the number of sections **140**A-N and **145**A-N. Body **110** may be connected to crown **115** by body fold **125**. Body fold **125** forms a pentadecagon or other polygonal shape depending on the number of sections **140**A-N (sections **140**A-B are not seen due to perspective) and **145**A-N (sections **145**A-B are not seen due to perspective). Body **110** may include multiple face sections **140**(A-N) and multiple fold sections **145**A-N. For example, body **110** may include fold section **145**B positioned between face section **140**B and fold section **140**C. In this manner, body **110** may alternate between face section **140**A, fold section **145**A, to face section **140**B, and fold section **145**B, to face section **140**C, and fold section **145**C, to face section **140**N, and fold section **145**N and so on. This process may extend circumferentially to where face sections **140**A-N and fold sections **145**A-N touch base **105** at a proximal end and touch crown **115** at a distal end. Proximal in this disclosure means closer to the center and distal in this disclosure means closer to an outside edge.

[0033] Face sections **140**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Similarly, fold sections **145**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Face sections **140**A-N may consist of a trapezoid-like shape with a curved bottom towards the proximal end where it meets base fold **120**. Alternatively, face section **140**A-N may be a trapezoid. Fold sections **145**A-N are smaller than half the size of face sections **140**A-N but also resemble a trapezoid-like shape with a curved side. The curved side is located on the proximal end of fold sections **145**A-N where it meets base fold **120**. Alternatively, fold section **145**A-N may be a trapezoid. The size and shape of face sections **140**A-N and **150**A-N and fold sections **145**A-N and **155**A-N may vary based on the desired shape of container **100**.

[0034] Folds **120**, **125**, **130**A-N, and **135**A-N may indicate where a bend in the material may occur or near where a bend is intended to be. Further, folds **120**, **125**, **130**A-N, and **135**A-N may represent a score in the material to facilitate bending in or near that location. Some bends may occur right along folds **120**, **125**, **130**A-N, and **135**A-N. In another example, a bend may occur near or around an area. For example, base fold **120** may be circularly shaped or circularly scored prior to the folding process to facilitate the fold. However, when the process of folding takes place the shape of the base fold **120** may resemble more polygonal than circular. Container **100** may be comprised of one or more of wood, paper, cardboard, metal, (aluminum, copper, brass), plastic, foam, rubber, or other foldable material known in the art.

[0035] Once folded container **100** may include multiple resting positions where material forces facilitated container **100** to remain in that position. For example, one position would be the flat position prior to fold initiation. A second example may be when container **100** is folded together such that a small polygonal-shaped opening is displayed at the top portion of the container and where the polygonal-shaped opening is positioned above body fold **125**. The third example may be when container **100** is folded together such that the polygonal-shaped opening is positioned below body fold **125**.

[0036] As container **100** is folded face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a counterclockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a clockwise direction. Facilitating the production of multiple resting positions. In an alternative embodiment, face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a clockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a counterclockwise direction.

[0037] Container **100** may be folded along base fold **120** towards base **105** such that base **105** may act as the base of container **100** and body **110** may act as the walls of container **100**. The angle of body **110** with respect to base **105** may vary. A greater angle between base **105** and body **110** creates a more gradual slope in the wall of container **100** wherein the distal end of body **110** extends away from a center point of container **100**. As the angle between body **110** and base **105** decreases the distal ends of body **110** come closer to a center point of container **100**. Accordingly, body **110** acts like the walls of container **100** become steeper. As a result, body **110** may be folded such that it may be substantially perpendicularly to the plane of base **105**. Substantially perpendicular in this circumstance means plus or minus 10 degrees.

[0038] Crown **115** may be circumferentially attached to body **110** and may fold inward at base fold **120** towards the center point of container **100**. Crown **115** may be folded at an angle such that the distal ends of crown **115** come together and extend over the top of base **105** to form a small polygonal-shaped opening. This opening, as seen in FIG. **5**, is positioned at the top portion of the container and where the opening is above (or higher than) a plane defined by body fold **125** creating a resting position for container **100**. Alternatively, the opening, positioned over base **105**, may be even with or below the plane defined by body fold **125** to create another resting position for container **100**.

[0039] To further facilitate the shape and resting positions of container **100**, container **100** may be folded along the one or more spoke folds **130**A-N and angled spoke folds **135**A-N. One or more spoke folds **130**A-N may fold towards the inside portion of container **100** on body **110**. At the same time, the same one or more spoke folds **130**A-N may be folded towards the outside portion of container **100** on the crown **115**. At the same time, one or more angled spoke folds **135**A-N may fold towards the outside portion of container **100** on body **110**. At the same time, the same one or more angled spoke folds **135**A-N may be folded towards the inside portion of container **100** on the crown **115**. In short, on body **110**, spoke folds **130**A-N may be folded in opposite directions as compared to angled spoke folds **135**A-N. On crown **115**, spoke folds **130**A-N may be folded in the opposite direction compared to spoke folds **130**A-N on body **110**. Similarly, on crown **115** angled spoke folds **135**A-N may be folded in opposite directions compared to angled spoke folds **135**A-N on body. As a result, following this pattern, regardless of the direction of the folds outward or inward, may create multiple resting positions for container **100**. The elasticity of the material of container **100** and the friction created by folds **120**, **125**, **130**A-N, and **135**A-N may facilitate the resting positions of container **100**.

[0040] FIG. **4** illustrates a perspective view of a rotational symmetric folding flanged container. Container **100** may include a base **105**, body **110**, and crown **115**. Base **105** may be circumferentially connected to body **110** by base fold **120**. The shape of base fold **120** forms a circle. Alternatively, base fold **120** may be a pentadecagon or other polygonal shape depending on the number of sections **140**A-N (sections **140**A-B are not seen due to perspective) and **145**A-N (sections **145**A-B are not seen due to perspective). Body **110** may be connected to crown **115** by body fold **125**. Body fold **125** forms a pentadecagon or other polygonal shape depending on the number of sections **140**A-N and **145**A-N. Body **110** may include multiple face sections **140**(A-N) and multiple fold sections **145**A-N. For example, body **110** may include fold section **145**B positioned between face section **140**B and fold section **140**C. In this manner, body **110** may alternate between face section **140**A, fold section **145**A, to face section **140**B, and fold section **145**B, to face section **140**C, and fold section **145**C, to face section **140**N, and fold section **145**N and so on. This process may extend circumferentially to where face sections **140**A-N and fold sections **145**A-N touch base **105** at a proximal end and touch crown **115** at a distal end. Proximal in this disclosure means closer to the center and distal in this disclosure means closer to an outside edge.

[0041] Face sections **140**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Similarly, fold sections **145**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Face sections **140**A-N may consist of a trapezoid-like shape with a curved bottom towards the proximal end where it meets base fold **120**. Alternatively, face section **140**A-N may be a trapezoid. Fold sections **145**A-N are smaller than half the size of face sections **140**A-N but also resemble a trapezoid-like shape with a curved side. The curved side is located on the proximal end of fold sections **145**A-N where it meets base fold **120**. Alternatively, fold section **145**A-N may be a trapezoid. The size and shape of face sections **140**A-N and **150**A-N and fold sections **145**A-N and **155**A-N may vary based on the desired shape of container **100**.

[0042] Folds **120**, **125**, **130**A-N, and **135**A-N may indicate where a bend in the material may occur or near where a bend is intended to be. Further, folds **120**, **125**, **130**A-N, and **135**A-N may represent a score in the material to facilitate bending in or near that location. Some bends may occur right along folds **120**, **125**, **130**A-N, and **135**A-N. In another example, a bend may occur near or around an area. For example, base fold **120** may be circularly shaped or circularly scored prior to the folding process to facilitate the fold. However, when the process of folding takes place the shape of the base fold **120** may resemble more polygonal than circular. Container **100** may be comprised of one or more of wood, paper, cardboard, metal, (aluminum, copper, brass), plastic, foam, rubber, or other foldable material known in the art.

[0043] Once folded container **100** may include multiple resting positions where material forces facilitated container **100** to remain in that position. For example, one position would be the flat position prior to fold initiation. A second example may be when container **100** is folded together such that a small polygonal-shaped opening is displayed at the top portion of the container and where the polygonal-shaped opening is positioned above body fold **125**. The third example may be when container **100** is folded together such that the polygonal-shaped opening is positioned below body fold **125**.

[0044] As container **100** is folded face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a counterclockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a clockwise direction. Facilitating the production of multiple resting positions. In an alternative embodiment, face sections **140**A-N and fold sections **145**A-N of body **110** may be folded such that the distal ends of sections **140**A-N and **145**A-N are angled in a clockwise direction. In contrast, face sections **150**A-N and fold sections **155**A-N of crown **115** may be folded such that the distal ends of sections are angled in a counterclockwise direction.

[0045] Container **100** may be folded along base fold **120** towards base **105** such that base **105** may act as the base of container **100** and body **110** may act as the walls of container **100**. The angle of body **110** with respect to base **105** may vary. A greater angle between base **105** and body **110** creates a more gradual slope in the wall of container **100** wherein the distal end of body **110** extends away from a center point of container **100**. As the angle between body **110** and base **105** decreases the distal ends of body **110** come closer to a center point of container **100**. Accordingly, body **110** acts like the walls of container **100** become steeper. As a result, body **110** may be folded such that it may be substantially perpendicularly to the plane of base **105**. Substantially perpendicular in this circumstance means plus or minus 10 degrees.

[0046] Crown **115** may be circumferentially attached to body **110** and may fold inward at base fold **120** towards the center point of container **100**. Crown **115** may be folded at an angle such that the distal ends of crown **115** come together and extend over the top of base **105** to form a small polygonal-shaped opening. This opening, as seen in FIG. **5**, is positioned at the top portion of the container and where the opening is above (or higher than) a plane defined by body fold **125** creating a resting position for container **100**. Alternatively, the opening, positioned over base **105**, may be even with or below the plane defined by body fold **125** to create another resting position for container **100**.

[0047] To further facilitate the shape and resting positions of container **100**, container **100** may be folded along the one or more spoke folds **130**A-N and angled spoke folds **135**A-N. One or more spoke folds **130**A-N may fold towards the inside portion of container **100** on body **110**. At the same time, the same one or more spoke folds **130**A-N may be folded towards the outside portion of container **100** on the crown **115**. At the same time, one or more angled spoke folds **135**A-N may fold towards the outside portion of container **100** on body **110**. At the same time, the same one or more angled spoke folds **135**A-N may be folded towards the inside portion of container **100** on the crown **115**. In short, on body **110**, spoke folds **130**A-N may be folded in opposite directions as compared to angled spoke folds **135**A-N. On crown **115**, spoke folds **130**A-N may be folded in the opposite direction compared to spoke folds **130**A-N on body **110**. Similarly, on crown **115** angled spoke folds **135**A-N may be folded in opposite directions compared to angled spoke folds **135**A-N on body. As a result, following this pattern, regardless of the direction of the folds outward or inward, may create multiple resting positions for container **100**. The elasticity of the material of container **100** and the friction created by folds **120**, **125**, **130**A-N, and **135**A-N may facilitate the resting positions of container **100**.

[0048] FIG. **5** illustrates a perspective view of a rotational symmetric folded flanged container. Container **100** may include a base **105**, body **110**, and crown **115**. Base **105** may be circumferentially connected to body **110** by base fold **120**. The shape of base fold **120** forms a circle. Alternatively, base fold **120** may be a pentadecagon or other polygonal shape depending on the number of sections **140**A-N and **145**A-N. Body **110** may be connected to crown **115** by body fold **125**. Body fold **125** forms a pentadecagon or other polygonal shape depending on the number of sections **140**A-N (not seen due to perspective) and **145**A-N (not seen due to perspective). Body **110** may include multiple face sections **140**(A-N) and multiple fold sections **145**A-N. For example, body **110** may include fold section **145**B positioned between face section **140**B and fold section **140**C. In this manner, body **110** may alternate between face section **140**A, fold section **145**A, to face section **140**B, and fold section **145**B, to face section **140**C, and fold section **145**C, to face section **140**N, and fold section **145**N and so on. This process may extend circumferentially to where face sections **140**A-N and fold sections **145**A-N touch base **105** at a proximal end and touch crown **115** at a distal end. Proximal in this disclosure means closer to the center and distal in this disclosure means closer to an outside edge.

[0049] Face sections **140**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Similarly, fold sections **145**A-N may be bordered by spoke folds **130**A-N on the first side, angled spoke fold **135**A-N on the second side, body fold **125** on the third side distally, and base fold **120** on the fourth side proximally. Face sections **140**A-N may consist of a trapezoid-like shape with a curved bottom towards the proximal end where it meets base fold **120**. Alternatively, face section **140**A-N may be a trapezoid. Fold sections **145**A-N are smaller than half the size of face sections **140**A-N but also resemble a trapezoid-like shape with a curved side. The curved side is located on the proximal end of fold sections **145**A-N where it meets base fold **120**. Alternatively, fold section **145**A-N may be a trapezoid. The size and shape of face sections **140**A-N and **150**A-N and fold sections **145**A-N and **155**A-N may vary based on the desired shape of container **100**.

[0050] Folds **120**, **125**, **130**A-N, and **135**A-N may indicate where a bend in the material may occur or near where a bend is intended to be. Further, folds **120**, **125**, **130**A-N, and **135**A-N may represent a score in the material to facilitate bending in or near that location. Some bends may occur right along folds **120**, **125**, **130**A-N, and **135**A-N. In another example, a bend may occur near or around an area. For example, base fold **120** may be circularly shaped or circularly scored prior to the folding process to facilitate the fold. However, when the process of folding takes place the shape of the base fold **120** may resemble more polygonal than circular. Container **100** may be comprised of one or more of wood, paper, cardboard, metal, (aluminum, copper, brass), plastic, foam, rubber, or other foldable material known in the art.

[0051] Once folded container **100** may include multiple resting positions where material forces facilitated container **100** to remain in that position. For example, one position would be the flat position prior to fold initiation. A second example may be when container **100** is folded together such that a small polygonal-shaped opening **160** is displayed at the top portion of the container and where polygonal-shaped opening **160** is positioned above body fold **125**. The third example may be when container **100** is folded together such that polygonal-shaped opening **160** is positioned below body fold **125**.

[0052] Container **100** may be folded along base fold **120** towards base **105** such that base **105** may act as the base of container **100** and body **110** may act as the walls of container **100**. The angle of body **110** with respect to base **105** may vary. A greater angle between base **105** and body **110** creates a more gradual slope in the wall of container **100** wherein the distal end of body **110** extends away from a center point of container **100**. As the angle between body **110** and base **105** decreases the distal ends of body **110** come closer to a center point of container **100**. Accordingly, body **110** acts like the walls of container **100** become steeper. As a result, body **110** may be folded such that it may be substantially perpendicularly to the plane of base **105**. Substantially perpendicular in this circumstance means plus or minus 10 degrees.

[0053] Crown **115** may be circumferentially attached to body **110** and may fold inward at base fold **120** towards the center point of container **100**. Crown **115** may be folded at an angle such that the distal ends of crown **115** come together and extend over the top of base **105** to form opening **160**. Opening **160** may be positioned at the top portion of the container and where opening **160** is above (or higher than) a plane defined by body fold **125** creating a resting position for container **100**. Alternatively, opening **160** may be positioned over base **105**, may be even with or below the plane defined by body fold **125** to create another resting position for container **100**.

[0054] To further facilitate the shape and resting positions of container **100**, container **100** may be folded along the one or more spoke folds **130**A-N and angled spoke folds **135**A-N. One or more spoke folds **130**A-N may fold towards the inside portion of container **100** on body **110**. At the same time, the same one or more spoke folds **130**A-N may be folded towards the outside portion of container **100** on the crown **115**. At the same time, one or more angled spoke folds **135**A-N may fold towards the outside portion of container **100** on body **110**. At the same time, the same one or more angled spoke folds **135**A-N may be folded towards the inside portion of container **100** on the crown **115**. In short, on body **110**, spoke folds **130**A-N may be folded in opposite directions as compared to angled spoke folds **135**A-N. On crown **115**, spoke folds **130**A-N may be folded in the opposite direction compared to spoke folds **130**A-N on body **110**. Similarly, on crown **115** angled spoke folds **135**A-N may be folded in opposite directions compared to angled spoke folds **135**A-N on body. As a result, following this pattern, regardless of the direction of the folds outward or inward, may create multiple resting positions for container **100**. The elasticity of the material of container **100** and the friction created by folds **120**, **125**, **130**A-N, and **135**A-N may facilitate the resting positions of container **100**.

[0055] Further, although specific implementations of the disclosure have been described and illustrated, the disclosure is not to be limited to the specific forms or arrangements of parts so described and illustrated. The scope of the disclosure is to be defined by the claims appended hereto, any future claims submitted here and in different applications, and their equivalents.

Claims

1. A container comprising: a base comprising: a base fold; a body circumferentially connected to the base at the base fold comprising: a body fold; one or more face sections; one or more fold sections; a crown circumferentially connected to the body at the body fold comprising: one or more face sections; and one or more fold sections.

2. The container of claim 1, further comprises: one or more spoke fold.

3. The container of claim 2, further comprises: one or more angled spoke folds.

4. The container of claim 3, wherein one of the face sections in the body is positioned next to one of the fold sections in the body.

5. The container of claim 4, wherein one of the face sections in the crown is positioned next to one of the fold sections in the crown.

6. The container of claim 5, wherein the one of the face sections in the body is positioned next to the face section in the crown.

7. The container of claim 6, wherein the one of the fold sections in the body is positioned next to the fold section in the crown.

8. The container of claim 7, wherein when the container is folded the spoke fold in the body is folded towards the center of the container.
9. The container of claim 8, wherein when the container is folded, the spoke fold in the crown is folded to the outside away from the center of the container.
10. The container of claim 6, wherein the container is folded, the angled spoke fold in the body is folded to the outside away from the center of the container.
11. The container of claim 10, wherein the container is folded the angled spoke fold in the body is folded towards the center of the container.
12. The container of claim 6, wherein the container is folded the crown includes an opening formed on the top of the container.
13. The container of claim 12, wherein the container is folded the opening formed on the top of the container is positioned above a plane defined by the body fold.
14. The container of claim 12, wherein the container is folded the opening formed on the top of the container is positioned in line with a plane defined by the body fold.
15. The container of claim 12, wherein the container is folded the opening formed on the top of the container is positioned below a plane defined by the body fold.
16. The container of claim 12, wherein the body is folded in an upward direction in comparison to the base such that the angle between the body and a plane defined by the base is greater than ninety degrees.
17. The container of claim 12, wherein the body is folded in an upward direction in comparison to the base such that the angle between the body and a plane defined by the base is substantially ninety degrees.
18. The container of claim 12, wherein the face sections and the fold sections of the body are folded such that the distal ends of the fold sections and the face sections are angled in a counterclockwise direction.
19. The container of claim 18, wherein the face sections and the fold sections of the crown are folded such that the distal ends of the face sections and the fold sections are angled in a clockwise direction.
20. The container of claim 1 wherein the base fold is circular.